Edge AI Indoor Anomaly Sound Detection Using XIAO ESP32S3 Sense and Edge Impulse

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BACKGROUND

- Indoor sound detection for safety and monitoring
- Cloud-based systems have latency and privacy issues
- Edge AI enables real-time, on-device detection

OBJECTIVE

- Real-time anomaly detection (Normal vs Abnormal)
- On-device data collection (ESP32 mic)
- Lightweight model using Edge Impulse

METHODOLOGY

Device: XIAO ESP32S3 Sense
Data: mic_value, 200 Hz, 3-second windows
Classes: Normal and Abnormal
Features: Spectrograms
Model: Lightweight CNN in Edge Impulse

RESULTS

Best Model Performance

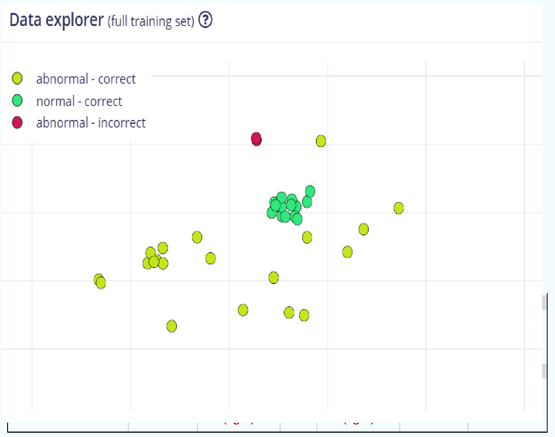
Validation accuracy: **87.5%**
Weighted precision: **0.90**
Weighted recall: **0.88**
Weighted F1-score: **0.87**

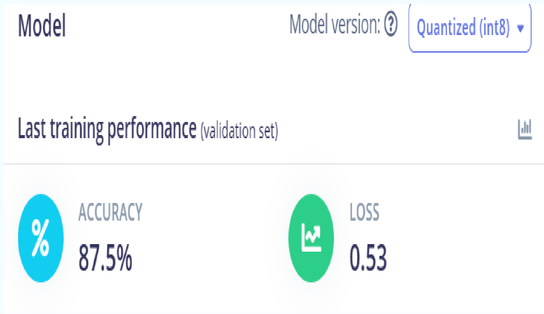
Confusion Matrix Summary

- Normal events correctly classified: **100%**
- Abnormal events correctly classified: **75%**
- Some abnormal samples were misclassified as normal

	ABNORMAL	NORMAL
ABNORMAL	75%	25%
NORMAL	0%	100%
F1 SCORE	0.86	0.89

LIVE CLASSIFICATION & MODEL INSIGHTS



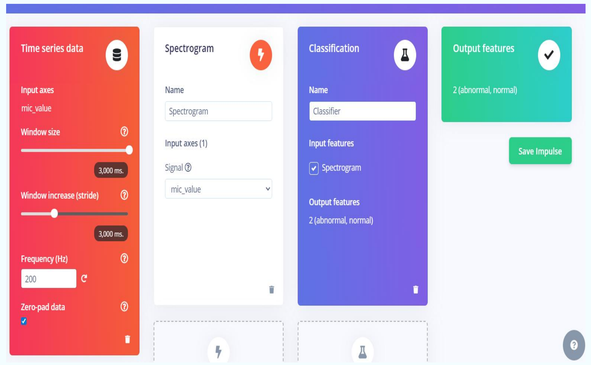


Live Performance Observations

- *Model separates normal and abnormal clusters effectively*
- *Strong spike events are classified more reliably*
- *Some overlap remains for low-intensity abnormal signals*

SYSTEM DESIGN & WORKFLOW

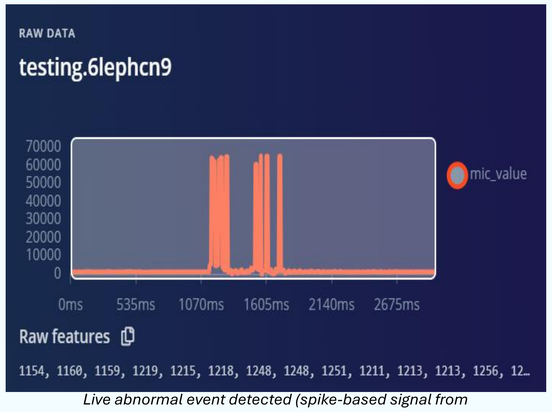
- Sound captured using XIAO ESP32S3 Sense3
- -second windows sampled at 200 Hz
- Data labeled as Normal or Abnormal
- Spectrogram + CNN used for classification
- Output: Normal / Abnormal



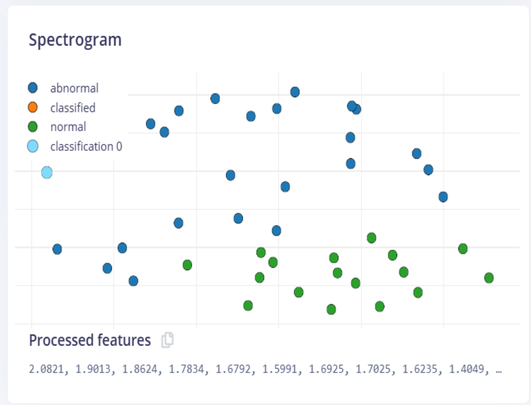
DEPLOYMENT & EDGE PERFORMANCE

Deployment & Edge Performance

- Deployed using Edge Impulse
- Quantized for efficient edge inference
- Runs directly on XIAO ESP32S3 Sense
- Inference time: 3 ms
- RAM usage: 12.7 KB
- Flash usage: 45.9 KB



Live abnormal event detected (spike-based signal from microphone)



CONCLUSIONS

- ***Edge-based abnormal sound detection was successfully demonstrated***
- ***The model achieved 87.5% validation accuracy***
- ***Sharp spike-like sounds were detected reliably in real time***

FUTURE WORK

- ***Collect more diverse abnormal sound samples***
- ***Improve abnormal-event recall***
- ***Compare mic_value and raw-audio classification***